Investigating the Effect of Simulated IMU Sensor Realism on Joint Angle Estimation Using Inverse Kinematics

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***Abstract*— Machine-learning-based estimation of joint kinematics from inertial measurement unit (IMU) data is limited by the availability of small training datasets and by discrepancies between idealized simulated sensor signals and real-world measurements affected by sensor imperfections. This project investigates how simulation-based IMU data augmentation with different levels of sensor realism influences joint angle estimation accuracy during walking. Using publicly available OpenSim musculoskeletal models, inverse kinematics (IK) results from a walking trial were used to generate virtual IMU signals, such that all IMU data in this study were simulated rather than recorded from physical sensors. Three datasets with increasing realism were defined: (D1) clean virtual IMU data without noise or bias, (D2) data with deterministic imperfections including scale errors, axis misalignment, and constant bias, and (D3) data with additional stochastic noise and drift. A sliding-window multi-output regression model was trained to predict right-leg hip flexion, knee flexion, and ankle plantarflexion/dorsiflexion, and performance was evaluated using root mean squared error (RMSE) on a temporally held-out test set. Training on clean data (D1) yielded an overall RMSE of 4.74°, while including deterministic sensor imperfections (D1+D2) reduced the overall RMSE to 4.44°, corresponding to a 6.4% improvement, with notable reductions for knee (7.29° → 6.70°, +8.2%) and ankle (2.38° → 2.24°, +5.9%) joint angles. Further augmentation with stochastic noise and drift (D1+D2+D3) resulted in only marginal overall improvement (4.72°, +0.59%) and mixed joint-specific effects. These results highlight that moderate, controlled sensor realism improves generalization, whereas excessive augmentation provides diminishing returns when using simulation-based IMU data for learning-based joint kinematics estimation.**

I. Introduction

Quantitative analysis of human gait is a central topic in biomechanics, with applications in clinical diagnostics, rehabilitation, sports science, and the development of assistive technologies such as prostheses, orthoses, and exoskeletons. In particular, lower-limb joint kinematics—most notably hip, knee, and ankle joint angles—play a key role, as they directly reflect movement coordination and joint loading during locomotion. Reliable estimation of these quantities is therefore essential in both research and applied settings.

Traditionally, joint kinematics have been obtained using laboratory-based measurement systems, such as optical motion capture (OMC) combined with force plates and musculoskeletal modeling. While these systems are considered the gold standard due to their high accuracy and temporal resolution, they suffer from several practical limitations. Motion capture laboratories require expensive equipment, extensive calibration procedures, and trained personnel, and measurements are restricted to controlled environments and short recording durations. As a result, their applicability to long-term or real-world gait monitoring is limited.

To address these limitations, wearable sensing technologies—particularly inertial measurement units (IMUs)—have gained increasing attention in recent years. IMUs, consisting of accelerometers and gyroscopes, are portable, cost-effective, and easy to deploy, making them well suited for ambulatory gait analysis, at-home rehabilitation, and real-world sports applications. Consequently, IMU-based gait analysis has become an active area of research [4].

Despite their advantages, estimating joint kinematics from IMU data remains challenging. IMU signals are affected by various error sources, including sensor noise, drift, constant bias, axis misalignment, and variability in sensor placement. Moreover, IMUs measure linear accelerations and angular velocities rather than joint angles directly, resulting in a complex, nonlinear mapping that varies across individuals. These challenges complicate both physics-based and data-driven approaches to IMU-based gait analysis.

Physics-based methods typically combine inertial data with biomechanical models and optimization or filtering techniques to reconstruct joint kinematics. Although such approaches benefit from physical interpretability and model-based constraints, they often require computationally expensive optimization procedures and can be sensitive to modeling assumptions, such as simplified joint representations or rigid-body approximations. While machine learning–based approaches shift much of the computational cost to the training phase, both deep learning models and simulation-based data generation pipelines can also involve substantial computational effort, particularly when large datasets or complex musculoskeletal simulations are used.

As an alternative, machine learning–based approaches have been increasingly explored for estimating biomechanical variables from IMU data. By learning a direct mapping between IMU signals and joint kinematics from labeled datasets, supervised learning models can capture complex temporal relationships in inertial measurements. In recent years, deep learning architectures have demonstrated promising performance in this domain; however, their success strongly depends on the availability of large, diverse, and well-annotated training datasets [1].

The acquisition of such datasets represents a major bottleneck in IMU-based gait analysis. Collecting synchronized IMU and motion capture data is time-consuming and costly, which typically limits dataset size, subject diversity, and experimental conditions. As a result, models trained on small laboratory datasets often exhibit reduced generalization when applied to new subjects, walking speeds, or sensor characteristics.

Simulation-based data generation has therefore emerged as a promising strategy to mitigate data scarcity. Using musculoskeletal simulations, joint kinematics derived from

motion capture can be used to generate virtual IMU signals by placing simulated sensors on model segments [2], [3]. This approach enables the generation of large datasets under controlled conditions, while allowing precise manipulation of sensor placement, orientation, and signal characteristics.

A prominent example of this strategy is the work by Dorschky et al. [1], who proposed a framework for estimating sagittal-plane gait biomechanics from inertial sensor data using convolutional neural networks trained on both measured and simulated IMU signals. Their results showed that augmenting measured data with simulated data can improve joint angle estimation accuracy, while also highlighting that increasing the amount of simulated data does not necessarily lead to continuous performance gains. This finding emphasizes the importance of balancing data quantity with sensor realism.

Motivated by these observations, the present project investigates the role of sensor realism in simulation-based IMU data generation. Publicly available OpenSim musculoskeletal models are used to generate motion data from inverse kinematics (IK) results of a walking trial. All IMU signals in this study are virtual and computed directly from IK kinematics, allowing precise control over sensor characteristics.

To keep the project scope manageable, three datasets with increasing levels of complexity are defined. Dataset D1 consists of clean virtual IMU signals without noise, bias, or drift and serves as a reference. Dataset D2 introduces deterministic sensor imperfections, such as scale errors, axis misalignment, and constant bias. Dataset D3 further augments the signals with stochastic noise and drift to approximate more realistic IMU behavior. By comparing machine learning models trained on these datasets, this project aims to analyze how different levels of simulated sensor realism affect joint angle estimation performance.

Rather than aiming for state-of-the-art accuracy, the primary objective of this work is methodological. By systematically varying sensor realism and evaluating performance using standardized error metrics, the project provides insights into the trade-offs involved in simulation-based data augmentation for IMU-driven gait analysis.

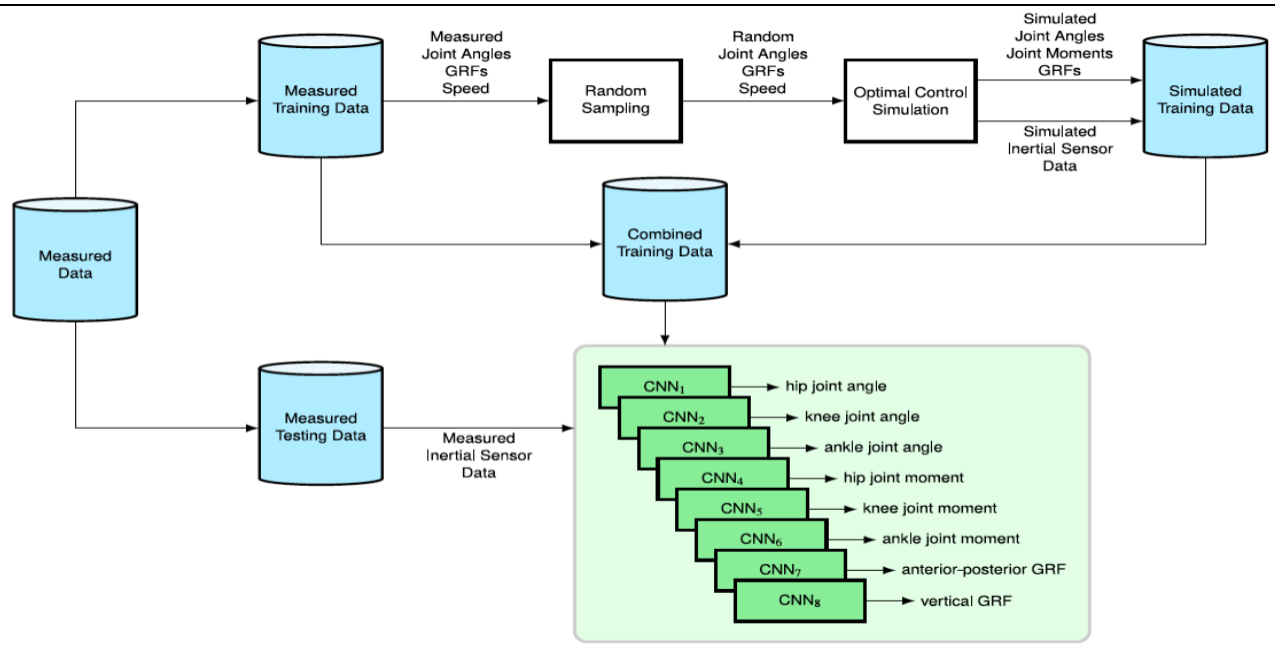

Figure 1: Overview of a simulation-based pipeline for generating virtual inertial sensor data and training machine learning models for gait analysis. The figure is adapted from Dorschky et al. [1]. and illustrates the general concept of combining biomechanical simulations with inertial sensor data for supervised learning. The present project implements a simplified variant of this pipeline.

II. Methods

The workflow starts from inverse kinematics (IK) motion data obtained from a walking trial recorded using optical motion capture and processed with a musculoskeletal model. These IK results were generated using a publicly available OpenSim full-body gait model from the OpenSim model repository. From the IK-derived kinematics, virtual inertial measurement unit (IMU) signals were computed by extracting segment orientations, angular velocities, and linear accelerations at selected body locations. Based on these signals, multiple datasets with increasing levels of sensor realism were generated and subsequently used to train and evaluate regression models for estimating lower-limb joint angles.

B. Musculoskeletal Model and Inverse Kinematics Data

All biomechanical data in this project are derived from simulations using the OpenSim software framework. A publicly available full-body gait model from the OpenSim model repository was applied to a walking trial for which inverse kinematics results were available. The inverse kinematics motion file contains time-resolved joint angle trajectories obtained from optical motion capture data and serves as the biomechanical ground truth throughout the study.

The analysis focuses on sagittal-plane joint kinematics of the right lower limb, specifically hip flexion, knee flexion, and ankle plantarflexion/dorsiflexion. These joints were selected because they are central to gait analysis and commonly investigated in IMU-based biomechanics. The IK motion data

were processed frame by frame, and the corresponding model states were applied to the musculoskeletal model to obtain segment positions and orientations in the global reference frame.

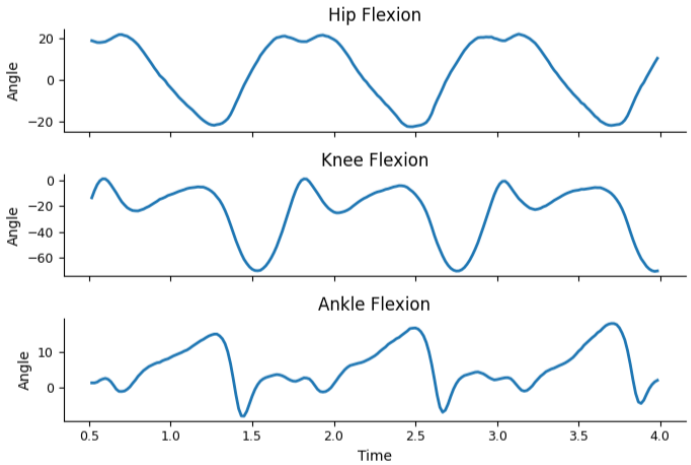

Figure 2: Inverse kinematics–derived sagittal-plane right hip, knee, and ankle joint angle trajectories from the walking trial used in this project, serving as ground-truth reference signals.

C. Virtual IMU Signal Generation

C.1 Virtual Sensor Placement and Kinematic Quantities

Virtual IMU signals were generated directly from the musculoskeletal model kinematics. Instead of physical sensors, virtual IMUs were attached to selected model segments, namely the pelvis and the right tibia. For each segment, the orientation with respect to the global reference frame was extracted at every time step. Although segment orientation is not measured directly by real IMUs, it is used here as an intermediate representation to derive angular velocity and linear acceleration signals.

C.2 Gyroscope and Accelerometer Signal Computation

Angular velocities were computed from consecutive orientation matrices using finite-difference approximations of rotation matrices, resulting in three-dimensional gyroscope signals. Linear accelerations were derived by differentiating segment origin positions twice with respect to time and expressing the resulting accelerations in the local sensor coordinate frame. Each virtual IMU therefore provided six channels: three angular velocity components and three linear acceleration components.

This simulation-based approach allows precise control over sensor placement and signal generation while avoiding experimental issues such as sensor misplacement, synchronization errors, or missing data. As a result, the generated IMU signals represent an idealized reference case of inertial measurements derived solely from kinematics, which also highlights a known limitation compared to real-world IMU data.

D. Dataset Construction and Sensor Realism Levels

To investigate the influence of sensor realism on model performance, three datasets with increasing complexity were constructed from the same underlying virtual IMU signals. All datasets share identical target variables, namely the joint angles extracted from the inverse kinematics results and differ only in the characteristics of the input IMU signals.

Dataset D1 consists of clean virtual IMU signals generated directly from the kinematics without any added imperfections. These signals contain no noise, bias, drift, scale error, or axis misalignment and therefore represent an idealized reference case.

In dataset D2, deterministic sensor imperfections were introduced to the IMU signals without adding stochastic noise. Constant biases, axis-dependent scale errors, and small rotational misalignments about the principal sensor axes were applied to both gyroscope and accelerometer channels to simulate systematic sensor errors commonly arising from manufacturing tolerances and imperfect sensor mounting.

In dataset D3, stochastic effects were added on top of the deterministic imperfections present in D2. These effects include additive Gaussian white noise and random-walk drift applied independently to each sensor axis. Noise magnitudes were chosen to be small relative to the signal amplitude and consistent with typical IMU noise characteristics reported in the literature, while drift was modeled as a low-frequency random walk to approximate long-term sensor instability. This dataset provides a more realistic, yet still controlled, approximation of IMU behavior.

Importantly, for all three datasets the target joint angle trajectories remain unchanged. Thus, any performance differences observed across the datasets can be attributed solely to differences in sensor realism rather than differences in biomechanical ground truth.

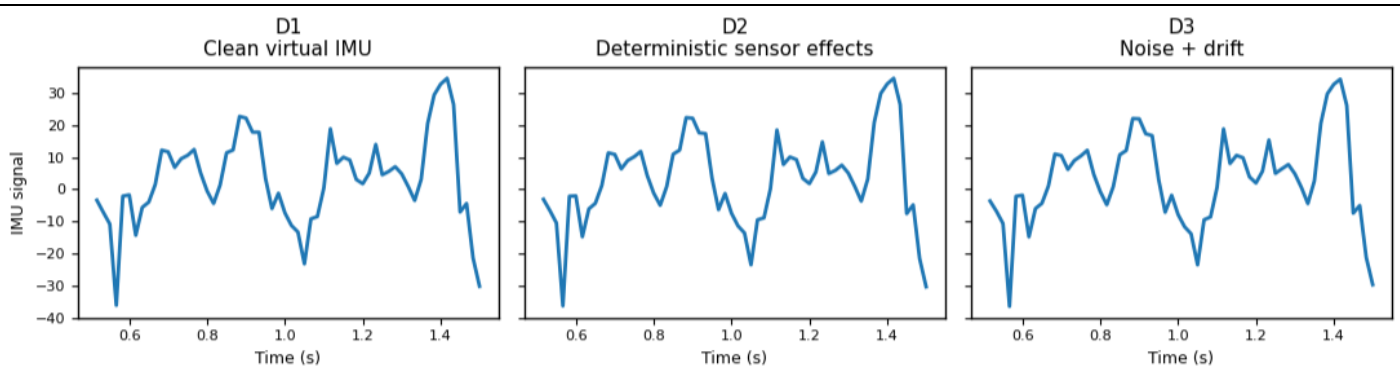

Figure 3: Representative IMU signal segments from the datasets used in this project, showing clean virtual data (D1), deterministic sensor imperfections (D2), and additional stochastic noise and drift (D3).

E. Feature Representation and Temporal Windowing

The IMU signals from all sensors have been integrated into a single multivariate time series, where, at every time step, all gyroscope and accelerometer channels are represented. To capture temporal dependencies in the gait dynamics, a sliding-gate approach was adopted. For each prediction, a fixed-length temporal window of IMU samples was collapsed into a feature vector, and the joint angles at the last time step of the window remained the corresponding targets.

This window-based representation allows for regressive modeling with a short-term temporal context being incorporated without the models having to explicitly encode sequential dependencies. The window length was chosen to approximately cover a meaningful portion of the gait cycle while maintaining reasonable feature dimensionality.

F. Machine Learning Models and Training Procedure

Joint angle estimation was formulated as a supervised multi-output regression problem. During development, several regression models were considered including ridge regression, random forest regression, and gradient boosting. These final experiments focus on the multilayer perceptron (MLP) regressor because it can effectively model the nonlinear relationships that exist between the IMU signals and joint kinematics.

The MLP was trained with input features that were standardized and optimized using the Adam optimizer, incorporating early stopping based on validation performance to lessen the risk of overfitting. The same model architecture and hyperparameters were enforced throughout all training schemes to make for a fair comparison of the datasets.

Training comprised three regimes: one utilizing only the clean dataset D1, the other using a mixture of D1 and D2, and one with a combination of D1, D2, and D3. In all instances, the test set consisted solely of clean data so as to assess generalization to idealized kinematics.

G. Evaluation Protocol and Performance Metrics

The held-out test set chronologically recorded with respect to the gait sequence preserves temporal structure of the gait sequence. Predicted accuracy was measured by calculating RMSE for each joint angle, where the RMSE value was then averaged across all joints.

In any case, besides absolute error values, relative performance improvement was estimated by noting RMSE values obtained under varying training regimes. This allowed assessing, directly, whether sensor augmentation during training provides generalization improvement.

H. Relation to Previous Work

Inspired substantially by earlier simulation-based studies, in particular that of Dorschky et al. [1], and colleagues, the methodological structure of this project does have however, introduce many simplifications. Differently from when either measured and simulated data are combined or optimal control simulations are used, this project relies solely on inverse kinematics-derived motion data and concentrates on the joint angle estimation. This design philosophy enables an even clearer isolation of the effects of sensor realism and supports a more interpretable analysis of simulation-based data augmentation.

III. Results

A. Reference Joint Kinematics from Inverse Kinematics

In this research work, the biomechanical ground truth is defined with respect to sagittal-plane joint angle trajectories as extracted from inverse kinematics (IK) simulation in OpenSim. Time-normalized joint angle profiles corresponding to right hip flexion, knee flexion, ankle plantarflexion/dorsiflexion over one walking gait cycle are shown in **Figure 3**. These trajectories demonstrate the well-established characteristics of gait patterns described in the biomechanics literature and are the reference targets for all training and testing phases. Notably, all datasets were generated using the same IK-derived joint angle trajectories and thus any differences in performance are solely attributed to differences in IMU signal characteristics.

B. Baseline performance using clean virtual IMU data (D1)

The model was first trained exclusively on the clean virtual IMU dataset (D1), which serves as the baseline condition. Performance was evaluated on a temporally held-out test set using the root mean square error (RMSE) between predicted and reference joint angles. As summarized in **Table 1**, the baseline RMSE amounted to **2.94°** for hip flexion, **7.29°** for knee flexion, and **2.38°** for ankle plantarflexion/dorsiflexion, resulting in an overall average RMSE of **4.74°** across all joints. These values provide the reference against which the effects of increasing sensor realism are assessed in subsequent experiments.

In addition to the scalar error metrics, **Figure 4** shows the time-series comparison between predicted and ground-truth joint angles over the gait cycle, illustrating phase-dependent errors and systematic bias that are not fully captured by RMSE values alone.

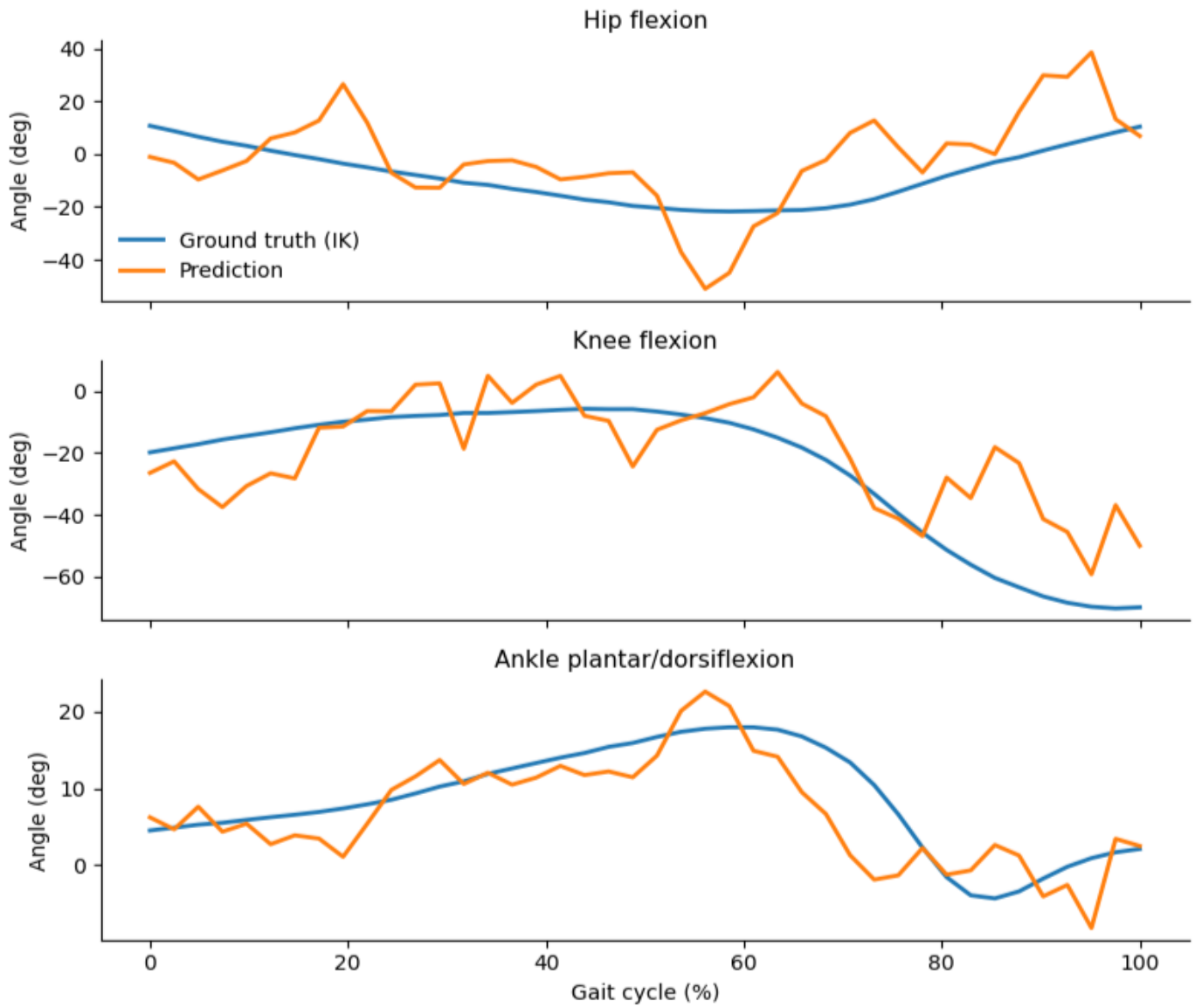

Figure 4: Time-normalized joint angle trajectories comparing ground-truth values from inverse kinematics (IK) and model predictions for right hip, knee, and ankle joints over one gait cycle. The plots qualitatively illustrate phase-dependent errors and systematic bias not captured by RMSE values alone.

C. Deterministic Imperfection Effects of the Sensors (D1+D2)

To analyze the influence of systematic sensor errors, the model was trained on a combined dataset consisting of clean data (D1) and data with deterministic imperfections (D2). Under this training regime, the RMSE for hip flexion slightly increased to **3.05°**, whereas the errors for knee flexion and ankle plantarflexion/dorsiflexion decreased to **6.70°** and **2.24°**, respectively. The resulting overall RMSE was reduced to **4.44°**, corresponding to an improvement compared to the baseline. These results indicate that exposure to deterministic sensor imperfections during training can enhance robustness, particularly for knee and ankle joint angle estimation.

Table 1: Root mean squared error (RMSE) of right hip, knee, and ankle joint angle estimation for models trained on clean virtual IMU data (D1), deterministic sensor-imperfect data (D1+D2), and additional stochastic augmentation (D1+D2+D3). Lower RMSE values indicate better estimation accuracy.

Training data	Hip flexion (°)	Knee flexion (°)	Ankle plantar/dorsi (°)	Overall (°)
D1	2,94	7,29	2,38	4,74
D1 + D2	3,14	6,70	2,29	4,44
D1 + D2 + D3	3,01	7,28	2,36	4,72

D. The Effect of Stochastic Noise and Drift (D1 + D2 + D3)

In the final experiment, stochastic sensor effects were introduced during training by additionally including dataset D3, which contains white noise and random-walk drift on top of deterministic imperfections. This configuration yielded an RMSE of **2.86°** for hip flexion, **7.28°** for knee flexion, and **2.36°** for ankle plantarflexion/dorsiflexion, resulting in an overall RMSE of **4.72°**. While hip flexion error showed a small improvement relative to the baseline, knee and ankle errors increased compared to the D1 + D2 configuration. Overall, the inclusion of stochastic noise and drift led to only marginal performance gains, suggesting limited additional benefit under the current setup.

E. Relative performance improvements

To facilitate comparison across training regimes, relative RMSE improvements with respect to the baseline D1 condition are summarized in **Table 2**. A positive percentage indicates a reduction in RMSE (i.e., improved performance), whereas a negative value denotes an increased error relative to the baseline. Training with deterministic sensor imperfections (D1 + D2) resulted in an overall RMSE reduction of 6.4%, with notable improvements for knee flexion (8.2%) and ankle plantarflexion/dorsiflexion (5.9%), despite a small degradation in hip flexion performance (−3.7%). In contrast, the addition of stochastic noise and drift (D1 + D2 + D3) yielded only a minor overall improvement of 0.59%, with negligible changes across individual joints. These results indicate that deterministic sensor imperfections contribute more effectively to generalization improvements than stochastic noise and drift.

F. Summary of findings

It is evident from the overall results that when training considering deterministic errors introduced by the sensor itself, the most consistent drop in joint angle estimation error is recorded. Introduction of stochastic noise and drift does not yield any further systematic improvement, but induces extra variability that may obscure the findings. Hence, it is crucial to strike the right balance between realism and control as far as simulation-based IMU data augmentation pipelines are concerned.

Table 2: Relative RMSE change (%) with respect to the D1 baseline for hip flexion, knee flexion, and ankle plantarflexion/dorsiflexion. Positive values indicate a reduction in RMSE (improved performance), whereas negative values indicate increased error.

Training data	RMSE vs D1 (%)			
	Hip	Knee	Ankle	Overall
D1 + D2	−3.68	+8.15	+5.89	+6.36
D1 + D2 + D3	+2.54	+0.25	+0.75	+0.59

IV. Discussion

This project investigated how increasing levels of simulated IMU sensor realism affect joint angle estimation performance when training models on virtual data derived from inverse kinematics. Since all datasets shared identical target joint angle trajectories, any performance differences can be attributed solely to changes in the input IMU signals.

Training on clean virtual IMU data (D1) already resulted in reasonable estimation accuracy, demonstrating that inverse kinematics–derived joint angles provide a suitable ground truth for learning the IMU–kinematics relationship. However, when deterministic sensor imperfections were included during training (D1+D2), the overall RMSE decreased, with notable improvements for knee flexion and ankle plantarflexion/dorsiflexion. This indicates that exposing the model to systematic sensor effects such as bias, scale error, and axis misalignment improves robustness and reduces sensitivity to idealized signals.

In contrast, adding stochastic noise and drift (D1+D2+D3) led to only minor additional changes in performance. While hip flexion error showed a small reduction, the overall improvement relative to the baseline remained limited, suggesting diminishing returns from further increasing sensor realism under the present conditions. These findings are consistent with prior work, including Dorschky et al. [1], who reported that simulated data can improve joint angle estimation but that additional realism does not necessarily lead to continuous performance gains.

Overall, the results suggest that moderate, controlled sensor realism is sufficient to improve generalization, whereas excessive augmentation may not be beneficial when training data are limited. Although this study was restricted to a single walking trial and sagittal-plane joint kinematics, it provides clear methodological insight into the design of simulation-based IMU datasets for machine learning in biomechanics.

V. Summary and Outlook

This project investigated the effect of simulated IMU sensor realism on machine learning–based joint angle estimation using virtual data derived from inverse kinematics. Virtual IMU signals were generated from an OpenSim musculoskeletal model, and three datasets with increasing levels of sensor realism were constructed while keeping the biomechanical ground truth identical. This controlled setup enabled a direct assessment of how sensor imperfections influence model performance.

The results showed that training on clean virtual IMU data already yields reasonable joint angle estimation accuracy. Introducing deterministic sensor imperfections during training improved overall performance, particularly for knee flexion and ankle plantarflexion/dorsiflexion, whereas further augmentation with stochastic noise and drift resulted in only marginal additional benefits. These findings highlight that moderate and physically meaningful sensor realism is sufficient to improve generalization, while excessive augmentation may lead to diminishing returns.

While the present study focused on a single walking trial and sagittal-plane joint kinematics, the methodology provides a clear framework for analyzing simulation-based data augmentation strategies in IMU-driven biomechanics. Future work could extend this approach to multiple subjects, additional movement tasks, and real IMU measurements to evaluate transferability to experimental data. Moreover, incorporating more advanced temporal models and optimizing sensor simulation parameters may further improve robustness and generalization in real-world applications.

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